

ABHINAV GOEL

ROBOTICS & AUTONOMOUS SYSTEMS | ROS 2 | UAV | MANIPULATION | LEGGED ROBOTICS

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PROFESSIONAL SUMMARY

Robotics-focused ECE undergraduate specializing in autonomous mobile robots, UAV autonomy, robotic manipulation, and ROS 2 systems. Hands-on experience developing and simulating AMRs and multirotor UAVs using ROS 2, Nav2, SLAM, ArduPilot, MAVSDK, computer vision, and embedded hardware. Hands-on across robot modeling, sensor integration, perception, planning, control, and hardware deployment.

TECHNICAL SKILLS

Robotics & Autonomous Systems:	ROS 2 (Humble), Nav2, ros2_control, SLAM Toolbox, AMCL, TF2, URDF/Xacro, MoveIt 2, Kinematics, Trajectory Planning, Sensor Fusion, EKF, RANSAC
UAV & Flight Systems:	ArduPilot, PX4, Pixhawk, MAVLink, MAVSDK, DroneKit, QGroundControl, Mission Planner, ArduPilot SITL, PID Control
Computer Vision & Perception:	OpenCV, YOLOv8, YOLOv4-Tiny, ArUco, Visual Servoing, Pinhole Camera Model, ArUco Marker Geolocation, ORB-SLAM3
Embedded Systems & Electronics:	ESP32-S3, ESP32-CAM, Arduino, STM32, UART, I ² C, SPI, PWM, MPU9250/MPU6050, Encoder Motors, Barometer, NRF24L01, LoRa, ST7920 Display, Custom MOSFET ESC
Compute, Sensors & Hardware:	Raspberry Pi 4/5, NVIDIA Jetson Nano/AGX Xavier, RPLidar A1M8, LiDAR, Raspberry Pi Camera v3 NoIR
Simulation & Engineering Tools:	Gazebo, RViz2, MuJoCo, MATLAB/Simulink, Fusion 360, 3D Printing, TinkerCAD, Git, Linux (Ubuntu 22.04)
Programming Languages:	Python, C++

EDUCATION

PDPM Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur Aug 2024 – Aug 2028

Bachelor of Technology (B.Tech) — Electronics and Communication Engineering

- Relevant Coursework: Robotics (Kinematics, D-H Parameters, Trajectory Planning), Control Systems, Embedded Systems, Sensors & Actuators.
- Practical Labs: PUMA Robot Kinematics, Trajectory-Following Autonomous Mobile Robots.

LEADERSHIP & EXPERIENCE

Coordinator (March 2026 – Present) | Member (Aug 2024 – March 2026) — ERS Aug 2024 – Present

Electronics and Robotics Society (ERS) — PDPM IIITDM Jabalpur

- Elected Coordinator in March 2026; mentoring 50+ junior students in hands-on robotics, covering ROS 2, autonomous systems, embedded systems, and practical robot development.
- Active society member since August 2024; engineered multiple society robotics platforms including 5-DOF manipulators, differential drive AMRs, and bio-inspired snake robots.

Core Committee Member (May 2025 – Present) | Member (Aug 2024 – May 2025) Aug 2024 – Present

Aero Fabrication Club — PDPM IIITDM Jabalpur

- Elevated to Core Committee Member in May 2025; developed autonomous UAV systems, custom ESCs, and integrated flight electronics with Pixhawk controllers and MAVLink-based communication.
- Represented the institution in national UAV challenges (SAE Aerothon 2025, NIDAR 2026), leading autonomous mission planning, boustrophedon survey algorithms, and flight testing.

FEATURED ROBOTICS PROJECTS

e-Yantra KrishiCobot — IIT Bombay (e-YRC 2025-26) [GitHub] Aug 2025 – May 2026

- ROS2 Humble · LiDAR · IMU · OpenCV · ArUco · UR5 Arm · Visual Servoing
- Developed an autonomous agricultural robot combining LiDAR, ultrasonic sensors, IMU, and ROS 2 for collision-aware navigation through crop rows.
 - Implemented perception and fruit-identification pipelines using OpenCV and ArUco markers, integrating detected targets with the robot's manipulation workflow.
 - Developed UR5 pick-and-place routines using visual servoing, coordinating perception, arm motion, and target interaction for autonomous harvesting.
 - Achieved Top 10 among 400+ teams in e-Yantra Robotics Competition 2025-26.

WareOps — Autonomous Warehouse Auditing Robot & Active Vision System [GitHub Org] Dec 2025 – Aug 2026

- ROS2 Humble · Nav2 · AMCL · SLAM Toolbox · ros2_control · RPLidar A1M8 · OpenCV · PyZbar · Gazebo Ignition · ESP32-CAM
- Developed an end-to-end autonomous AMR pipeline for rack-to-rack warehouse navigation using ROS2, Nav2, AMCL, and SLAM Toolbox for localization, mapping, and path planning.
 - Designed a 2-DOF active-vision scanning mast with pan/tilt gimbal sweeps and dual QR-detection pipelines using OpenCV WeChat QR and PyZbar for multi-tier rack auditing.
 - Integrated simulation and hardware stacks, replicating the physical robot in Gazebo Ignition using URDF/Xacro and SDF with RPi 4, RPLidar A1M8, Arduino Nano/ros2_control, and ESP32-CAM Wi-Fi vision.

[Disaster Surveillance Drone — NIDAR 2026](#) [\[GitHub\]](#)

Jun 2025 - Jan 2026

Python · MAVSDK · ArduPilot SITL · MAVLink · QGroundControl · Mission Planner · YOLOv8

- Developed polygon-based autonomous survey planning using KML boundaries and optimized zigzag coverage paths, covering 75+ acres in under 20 minutes at 25 m altitude.
- Integrated YOLOv8 human detection with autonomous target response, including target-centered tracking, pause/resume behavior, geofencing, and Return-to-Launch (RTL) failsafe handling.
- Automated the complete UAV mission lifecycle using the MAVSDK Python API, from mission upload and arming through autonomous takeoff, survey execution, target detection, and RTL.

[Autonomous Drone Survey & Target Detection — SAE Aerothon 2025](#) [\[GitHub\]](#)

May 2025 - Nov 2025

Python · ROS2 · DroneKit · Pixhawk · ArduPilot · MAVLink · Gazebo · YOLOv4-Tiny · PID Control

- Developed an autonomous UAV system for aerial survey and target detection using ROS 2, ArduPilot, Pixhawk, and MAVLink, selected for national-level competition presentation.
- Implemented pinhole-camera-model-based pixel-to-GPS projection to estimate geographic target locations from aerial imagery.
- Developed a visual PID control loop for autonomous target centering and lock-on during hover operations.

[Differential Drive Robot — ROS2 Navigation & SLAM](#) [\[GitHub\]](#) (★ 7 Stars)

Jun 2025 - July 2025

C++ · Python · ROS2 Humble · Gazebo · Nav2 · SLAM Toolbox · AMCL · URDF/xacro · EKF

- Modeled and simulated a 4-wheeled differential-drive robot using URDF/Xacro, integrating robot kinematics, sensors, and Gazebo simulation.
- Implemented LiDAR-based 2D SLAM and localization using SLAM Toolbox and AMCL for autonomous navigation in GPS-denied indoor environments.
- Integrated Nav2 for autonomous path planning, obstacle avoidance, and goal navigation, with EKF-based sensor fusion for improved state estimation.

KEY ACHIEVEMENTS & COMPETITIONS

- **e-Yantra Robotics Competition 2025–26 (IIT Bombay):** Top 10 among 400+ teams ([Certificate](#))
- **DD Robocon 2025:** Stage-2 Qualification ([Certificate](#))
- **SAE Aerothon 2025:** Phase 2 Qualification ([Certificate](#))
- **NIDAR 2026:** National UAV Challenge ([Certificate](#))